

MediQore Full-Stack Implementation Roadmap

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Overview

MediQore is built in five phases over about 28 weeks: a 2-week foundation, your three module phases, and a 4-week hardening phase before final submission. Module 10 grows in every phase, so the supervisor portal always shows what the mobile app can already collect.

Every feature below is tagged with its scope ID (for example M5 FE-2 = Module 5, FE-2), so each task traces back to the final scope document. Week numbers are relative; map them onto your department's FYP-I and FYP-II evaluation dates.

Phase	Modules	Weeks	Lead (support)	Exit gate
0. Foundation	Shared setup, schema, auth and sync skeleton	1-2	Shared	App, API, DB and dashboard run end to end on one test record
1. Core field workflow	M1, M2, M3, M10 base	3-9	Zain Abbas (Zain Ali: backend, sync, web)	LHW registers a woman and records a visit offline; it syncs and appears on the dashboard
2. Intelligence and emergency	M4, M5, M6, M10 update	10-17	Zain Abbas: M4, M10. Zain Ali: M5, M6	Visit produces a risk result with Urdu explanation; emergency alert reaches supervisor by internet, SMS and call
3. Child health and campaigns	M7, M8, M9, M10 final	18-24	Zain Ali: M7, M8, M9. Zain Abbas: M10	All ten modules work; dashboard and reports cover every module

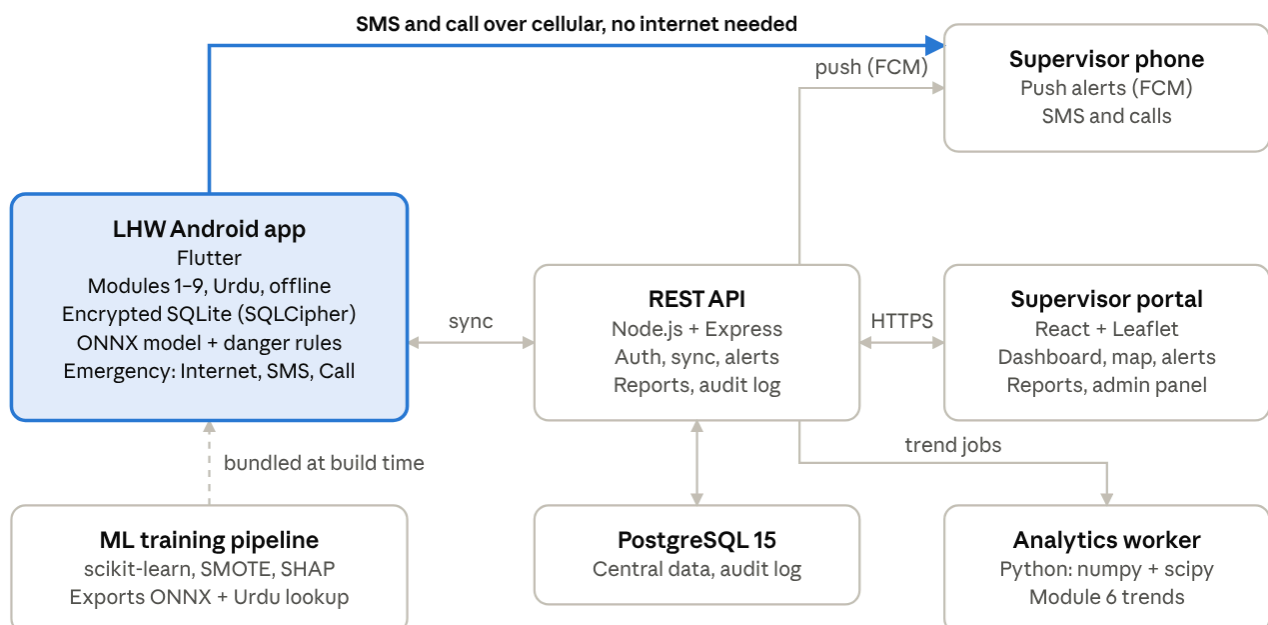
Phase	Modules	Weeks	Lead (support)	Exit gate
4. Hardening and delivery	Testing, performance, documentation	25–28	Shared	Full test pass on a low-spec Android device; final documentation and demo ready

Owners follow Table 4 of the scope document. Shared work (REST API, PostgreSQL schema, offline sync, integration testing, Figma, deployment) is split inside each phase as listed in the phase sections.

Architecture and conventions

The Flutter app owns the whole field workflow and never needs the server to do its job; the server only collects, analyses and reports. Settle the rules below in Phase 0, because changing them after Phase 1 means rewriting synced data.

The phone works alone; the server only collects, analyses and reports



MediQore architecture · 7 components

The highlighted path is the key design choice: an emergency SMS or call goes straight from the LHW's phone to the supervisor, with no server in between.

Component	Technology (from scope)	Responsibility
LHW mobile app	Flutter 3.x, Drift + sqflite_sqcipher, flutter_tts, onnxruntime, google_mlkit_text_recognition, fl_chart	Modules 1–9 in Urdu, fully offline, on-device AI and danger-sign rules, emergency alerts
REST API	Node.js 20 + Express, jsonwebtoken, Joi, pdfkit + ExcelJS	Auth, sync endpoints, conflict detection, alerts, reports, audit log
Database	PostgreSQL 15	Central store for all modules, audit log, sync sequence numbers
Supervisor and admin portal	React 18, Leaflet.js	Module 10: dashboard, map, alerts, reports, admin panel
ML and analytics	Python 3.11, scikit-learn, imbalanced-learn, SHAP, sklearn2onnx, numpy + scipy	Offline model training and export; backend trend analysis for Module 6
Push notifications	Firebase Cloud Messaging	Layer 1 emergency alerts (M5 FE-2)

Use one GitHub monorepo so the API contract, migrations and app change together:

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mediqore/
  mobile/  Flutter app (LHW role + supervisor alert role)
  api/     Node.js + Express REST API
  web/     React supervisor and admin portal
  ml/      Python training, SHAP lookup, ONNX export, analytics worker
  db/      PostgreSQL migrations and synthetic seed scripts
  docs/    OpenAPI contract, ERD, test reports

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Area	Rule
Record IDs	Every record made on the device gets a UUID v4; the server adds a sequence number when it accepts it. Never order records by device clock (LI-7).
Deletes and audit	Soft deletes only. Every create, edit, delete, referral and login writes an audit row with user and timestamp (M10 FE-3).

Area	Rule
API	REST under /api/v1, JSON, Joi validation on every request body, JWT access token plus refresh token, HTTPS only (M1 FE-2).
Units	Store one unit per vital: BP in mmHg, temperature in °C, blood sugar in mmol/L, weight in kg, MUAC in mm. Convert only at the model input or display layer.
Clinical rules	Danger-sign thresholds, EPI schedule, MUAC cut-offs and IMCI rules live in versioned JSON config files, never hard-coded, so clinical advisors can review them (M4 FE-4).
Urdu text	All labels in Flutter ARB localisation files, rendered in Jameel Noori Nastaleeq inside RTL Directionality; numeric vitals stay left to right.
Git workflow	main is always demo-ready; feature branches per FE (for example feature/m3-fe1-visit-form); every merge reviewed by the other member; GitHub Actions runs lint and tests.
Environments	Local PostgreSQL for development; one HTTPS staging server for supervisor reviews and demos.

Phase 0: Foundation (weeks 1–2)

Phase 0 builds the skeleton every module plugs into, so Phase 1 is spent on features rather than setup. Design the database for all ten modules now, even though most tables stay empty until later phases.

ID	Task	Scope link	Owner
P0-1	Create the monorepo, branch protection on main, and a GitHub Actions workflow that runs lint and unit tests	Tools: Git + GitHub	Shared
P0-2	Flutter project: Urdu ARB localisation, bundled Jameel Noori Nastaleeq font, RTL theme, and a small widget kit (large buttons, form fields, Green/Yellow/Red risk chips, offline status bar)	M3 FE-1	Zain Abbas

ID	Task	Scope link	Owner
P0-3	Express API skeleton: routes, controllers, services, Joi validation middleware, central error handler, request logging, OpenAPI file	M1 FE-2	Zain Ali
P0-4	PostgreSQL schema v1 with migrations for all core entities (see Data model) and a migration tool such as Knex or node-pg-migrate	Shared: DB schema	Zain Ali
P0-5	React portal skeleton: routing, login page, auth guard, layout with sidebar, empty Leaflet map component	M10	Zain Ali
P0-6	Offline sync skeleton: device outbox table, POST /sync/push and GET /sync/pull?since=, one record proven end to end	M3 FE-2	Shared
P0-7	Figma screens for Phase 1: login, LHW home, registration, visit form, dashboard home, admin lists	Mockups 1, 2, 6	Shared
P0-8	Synthetic data generator: districts, Union Councils, LHWs, households with GPS, pregnant women, visits	LI-10	Zain Ali
P0-9	Draft and submit the IEC application (due in Semester 7)	LI-10	Shared
P0-10	Start the ML track: download the UCI dataset, exploratory notebook, check duplicates and class balance	M4 FE-1	Zain Abbas
P0-11	Decide the open items in Risks and decisions (OTP channel, Urdu PDF method, Urdu voice source, model inputs)	All	Shared

Exit gate for Phase 0

- ☐ One test record created on the phone offline, synced, stored in PostgreSQL and visible on the React portal
- ☐ CI runs on every pull request
- ☐ Schema v1 and OpenAPI v1 reviewed by both members
- ☐ IEC application submitted

Phase 1: Core field workflow — M1, M2, M3, M10 base (weeks 3–9)

By the end of Phase 1 an LHW can log in, register a pregnant woman and record a visit with no internet, and the supervisor sees it on the portal after sync. Zain Abbas builds the mobile modules; Zain Ali builds the API, sync engine and most Phase 1 web screens, because his own modules start in Phase 2.

Module 1: LHW onboarding and access control

Task	Layer	Scope link	Owner
Admin creates an LHW with district, Union Council and area; system issues a unique LHW ID and credentials	API + Web	FE-1	Zain Ali
On first login the app downloads only that LHW's area data (her patient list)	Mobile + API	FE-1	Shared
JWT login with access and refresh tokens, login rate limiting, HTTPS only, server-side validation	API	FE-2	Zain Ali
OTP verification on first login and on a new device (channel decided in P0-11)	API + Mobile	FE-2	Zain Ali
Offline login: derive the SQLCipher key from the password with a slow key-derivation function (PBKDF2 or Argon2) and a stored salt; a correct password unlocks the local database	Mobile	FE-2, LI-8	Zain Abbas
Auto-lock after inactivity and session expiry; deactivated accounts are refused at next sync	Mobile + API	FE-2, FE-3	Shared
Admin area reassignment, activation and deactivation, password reset with a "sync before reset" warning	Web + API	FE-3, LI-8	Zain Ali

Module 2: Expecting woman registration

Task	Layer	Scope link	Owner
Registration form (name, age, husband's name, contact, address, pregnancy month, village) in Urdu with validation	Mobile	FE-1	Zain Abbas

Task	Layer	Scope link	Owner
Patient ID that is unique offline: LHW code plus a local counter, alongside the record UUID	Mobile	FE-1	Zain Abbas
Create the pregnancy file (one woman can have several pregnancies over time)	Mobile + DB	FE-1	Zain Abbas
Obstetric history: previous pregnancies, C-sections, stillbirths, known conditions; stored as the baseline risk profile	Mobile	FE-2	Zain Abbas
Household entity with GPS capture (geolocator package), linked to the woman; reused later by polio and child modules	Mobile + DB	FE-3	Zain Abbas
Area-based patient search and list sorting by village	Mobile	FE-3	Zain Abbas

Module 3: Field visit and vitals collection

Task	Layer	Scope link	Owner
Visit form: BP, weight, temperature, fetal movement, swelling, bleeding, fever, anaemia signs, urine symptoms, using checkboxes, dropdowns and large controls	Mobile	FE-1	Zain Abbas
Add pulse (heart rate) and optional blood sugar fields now, because the Phase 2 model needs them (see Risks)	Mobile	FE-1, M4 FE-1	Zain Abbas
Range checks to catch typing errors (for example systolic BP outside 60–250 asks for confirmation)	Mobile	FE-1	Zain Abbas
AES-256 encrypted local storage with Drift + sqlite_sqlcipher	Mobile	FE-2	Zain Abbas
Sync engine: outbox with retry, idempotent push by UUID, server sequence IDs, pull by cursor, background sync when online	Mobile + API	FE-2	Shared
Conflict detection for duplicate offline submissions; conflicted records flagged for supervisor review and kept in the audit log	API	FE-2, LI-7	Zain Ali

Task	Layer	Scope link	Owner
Voice guidance: on field focus, flutter_tts reads the Urdu label; mute toggle in settings	Mobile	FE-3, LI-6	Zain Abbas

Module 10 base: supervisor portal

Task	Layer	Scope link	Owner
Admin panel: district, tehsil, Union Council and area hierarchy; LHW and supervisor accounts; hospitals and referral centres; role permissions	Web + API	FE-3	Zain Ali
Audit log viewer with filters by user, action and date	Web + API	FE-3	Zain Ali
Dashboard cards: registered patients, visits this week, LHW visit counts, last login	Web + API	FE-1	Zain Ali
Leaflet map of registered households from Module 2 GPS, filter by district, Union Council, LHW and time period, auto-refresh every 5 minutes	Web	FE-1	Zain Abbas
Sync conflict review queue for supervisors	Web + API	M3 FE-2	Zain Ali

ML track (runs alongside Phase 1)

Zain Abbas trains the model now so Module 4 starts in Phase 2 with a ready ONNX file. Details are in Phase 2 under Module 4.

Exit gate for Phase 1

- ☐ LHW logs in online once, then works fully offline: login, registration, visit with voice guidance
- ☐ Records sync automatically when online; a duplicate submission is flagged, not overwritten
- ☐ Admin manages areas, accounts and hospitals; every action appears in the audit log
- ☐ Supervisor sees patients, visits, LHW activity and the household map
- ☐ Unit tests for form validation, sync and auth pass in CI

Phase 2: Intelligence and emergency — M4, M5, M6, M10 update (weeks 10–17)

By the end of Phase 2 every visit produces a risk result with an Urdu explanation, a critical case can reach the supervisor by internet, SMS or call, and each pregnancy has a full health record with trends. Zain Abbas owns Module 4 and the Module 10 update; Zain Ali owns Modules 5 and 6.

Module 4: AI-based maternal risk assessment

Task	Layer	Scope link	Owner
Clean the UCI data: remove duplicate rows, keep a stratified 20% hold-out test set untouched until the end	ML	FE-1	Zain Abbas
imbalanced-learn Pipeline with SMOTE applied inside each training fold only, so synthetic rows never leak into validation	ML	FE-1, BO-2	Zain Abbas
Compare Logistic Regression, Random Forest and Gradient Boosting with 5-fold stratified CV; report F1-macro, per-class F1, per-class recall and ROC-AUC	ML	FE-1, BO-2	Zain Abbas
Select the model: high-risk recall of at least 90% first, then best F1-macro	ML	BO-2	Zain Abbas
Write the PDHS 2017–18 feature-alignment note for the report	ML	FE-1, LI-2	Zain Abbas
Run SHAP on the chosen model, derive feature-threshold rules per class, map each to a reviewed Urdu phrase, export the JSON lookup	ML	FE-2	Zain Abbas
Export to ONNX with sklearn2onnx (plain probability output, no ZipMap); parity test that Python and ONNX agree on every test row	ML	FE-1	Zain Abbas
Run inference with onnxruntime after each visit; convert units to the dataset's (UCI body temperature is in °F); handle a missing blood sugar value as decided in P0-11	Mobile	FE-1	Zain Abbas
Benchmark inference time on the lowest-spec test phone (target under 1 second)	Mobile	FE-1	Zain Abbas

Task	Layer	Scope link	Owner
Result screen: colour-coded risk, Urdu explanation from the lookup, decision-support disclaimer	Mobile	FE-2, LI-5	Zain Abbas
Risk trend across visits: store each result, alert the LHW when the level or BP trend worsens across the last visits	Mobile	FE-3	Zain Abbas
Danger-sign rules engine read from JSON config; final level = higher of model and rules; Emergency hands off to Module 5; one unit test per rule	Mobile	FE-4, LI-12	Zain Abbas

Module 5: Emergency referral coordination

Task	Layer	Scope link	Owner
One-tap referral pre-filled with patient details, risk factors and the nearest hospital, found by GPS distance to the cached hospital list	Mobile	FE-1	Zain Ali
Emergency screen with three equal buttons (Internet, SMS, Call) and Send all; each button shows if it is usable now (connectivity_plus plus the phone's service state)	Mobile	FE-2	Zain Ali
Layer 1: alert to the API, Firebase Cloud Messaging push to the supervisor's phone, and a live alerts panel on the dashboard	Mobile + API + Web	FE-2	Zain Ali
Layer 2: pre-filled SMS sent from the LHW's phone (another_telephony, SEND_SMS permission) with sent and delivered callbacks	Mobile	FE-2, LI-4	Zain Ali
Layer 3: one-tap call to the supervisor and the secondary contact (flutter_phone_direct_caller, CALL_PHONE permission)	Mobile	FE-2	Zain Ali
Persistent emergency screen, Urdu voice announcement, protocol checklist, per-option status (Sent, Failed, Not available) with retry	Mobile	FE-4	Zain Ali
Save the alert record locally and sync it in the background when internet returns (workmanager); record time-to-escalation	Mobile + API	FE-4	Zain Ali

Task	Layer	Scope link	Owner
Acknowledgement: supervisor acknowledges in the app or dashboard; LHW can mark "supervisor reached" after a call or SMS reply	Mobile + Web + API	FE-5	Zain Ali
15-minute escalation on two sides: a server job for alerts the server knows about, and a device timer that prompts the LHW to contact the secondary contact for SMS-only or call-only alerts	Mobile + API	FE-5	Zain Ali
Referral outcome: attended or not, and outcome, visible to the supervisor	Mobile + Web	FE-3	Zain Ali

Module 6: Pregnancy journey, health records and ANC monitoring

Task	Layer	Scope link	Owner
ANC schedule generated from gestational age at registration; local reminders (flutter_local_notifications); missed visits flagged	Mobile	FE-1	Zain Ali
TT dose schedule plus iron and folic acid tracking, with a supplement compliance score on the patient profile	Mobile	FE-1	Zain Ali
Report scan: camera or gallery, grayscale and contrast preprocessing, ML Kit text recognition fully offline	Mobile	FE-2, LI-9	Zain Ali
Dart regex engine extracting BP, haemoglobin, blood glucose, temperature, weight and urine protein; high-confidence values pre-filled, others confirmed; manual entry fallback	Mobile	FE-2, LI-9	Zain Ali
Store the original report image (compressed, encrypted on device, uploaded on sync)	Mobile + API	FE-2	Zain Ali
Vital trend graphs on the device with fl_chart, built from local data so they work offline	Mobile	FE-3	Zain Ali
Python analytics worker (numpy + scipy) called by the API: linear-regression slope per vital, Z-score anomalies, weighted compliance score; results returned on next sync	ML + API	FE-3	Zain Ali

Task	Layer	Scope link	Owner
Bilingual PDF progress report: trend graphs, visit history, ANC compliance, doctor notes, Urdu summary	API	FE-3	Zain Ali

Module 10 update

Task	Layer	Scope link	Owner
Risk distribution chart (Green, Yellow, Red) and high-risk cluster layer on the map	Web + API	FE-1	Zain Abbas
Referral completion rate, overdue follow-ups and missed ANC visits per LHW	Web + API	FE-1	Zain Abbas
Emergency alerts panel: channel used, time-to-escalation, acknowledge button, unacknowledged alerts highlighted	Web + API	M5 FE-5	Zain Abbas
Admin: emergency escalation contacts per area (supervisor and secondary numbers)	Web + API	FE-3	Zain Abbas
Weekly and monthly PDF and Excel reports: maternal summary, high-risk list, referral completion, LHW activity, emergency alerts	API + Web	FE-2	Zain Abbas
Inactivity anomaly flag: weekly visits more than 2 standard deviations below the LHW's own mean, only after 4 weeks of history	API + Web	FE-4	Zain Abbas

Exit gate for Phase 2

- ☐ Chosen model meets at least 90% high-risk recall on the hold-out set, and ONNX parity test passes
- ☐ Every danger-sign rule forces an Emergency result in tests, even when the model says low risk
- ☐ Emergency alert sent by each of the three options in airplane mode (SMS and call) and online (internet)
- ☐ Unacknowledged alert escalates to the secondary contact after the set time
- ☐ OCR pre-fills values from a printed report; a handwritten report falls back to manual entry
- ☐ Dashboard shows risk, referrals, alerts and reports for maternal data

Phase 3: Child health and campaigns — M7, M8, M9, M10 final (weeks 18–24)

By the end of Phase 3 all ten modules work and the portal reports on every one of them. Start with a shared child registry in week 18, because Modules 7, 8 and 9 all depend on it and the scope does not assign it to a single module.

Shared child registry (week 18)

Task	Layer	Scope link	Owner
Household roster for every household in the LHW's area, not only those with a pregnant woman, reusing the Phase 1 household entity and GPS	Mobile + DB	M7 FE-1	Shared
Child record: name, date of birth, sex, household, optional link to the mother's pregnancy file	Mobile + DB	M7 FE-3, M8 FE-1	Shared
Child list per household with age shown in months, used by all three modules	Mobile	M7–M9	Shared

Module 7: Polio campaign field operations

Task	Layer	Scope link	Owner
Campaign rounds (name, dates, area) created on the portal and pulled to devices	Web + API + Mobile	FE-1	Zain Ali
House-to-house record: children under 5, children vaccinated, vaccine type (OPV), date; saved offline	Mobile	FE-1	Zain Ali
Refusal record with reason dropdown (religious concern, misinformation, past reaction, absent family) and an automatic revisit in the same round	Mobile	FE-2	Zain Ali
Revisit list for the LHW, sorted by distance from her current location	Mobile	FE-2	Zain Ali

Task	Layer	Scope link	Owner
Zero-dose check: any child under 5 with no OPV dose in any round goes on a priority list, on the device for her area and on the server for the whole district	Mobile + API	FE-3	Zain Ali

Module 8: Child immunisation and EPI management

Task	Layer	Scope link	Owner
EPI schedule config (BCG, OPV-0, Penta 1-3, PCV 1-3, Rota 1-2, IPV, MR) with due ages taken from the official Pakistan EPI schedule	Config	FE-1	Zain Ali
Personal vaccination timeline per child under 2 from date of birth; dose recording; birth-dose OPV-0 kept separate from campaign doses	Mobile	FE-1	Zain Ali
Defaulter alert on the LHW home screen when a dose passes its due date; supervisor notified after sync	Mobile + API	FE-2	Zain Ali
Coverage per antigen and per sub-area, with low-coverage sub-areas highlighted	API + Web	FE-3	Zain Ali

Module 9: Child nutrition and growth screening

Task	Layer	Scope link	Owner
MUAC entry with WHO classification: below 115 mm SAM, 115-125 mm MAM, above 125 mm Normal; colour-coded Urdu result; age check for 6-59 months	Mobile	FE-1	Zain Ali
Weight-for-age and height-for-age Z-scores from bundled WHO Child Growth Standards tables; stunting, wasting and underweight flags	Mobile	FE-1	Zain Ali
Unit tests that check Z-score output against WHO reference values	Mobile	FE-1	Zain Ali

Task	Layer	Scope link	Owner
SAM referral to the nearest Nutrition Rehabilitation Centre, attendance tracking and weight recovery across follow-ups	Mobile + Web	FE-2	Zain Ali
IMCI checklist: respiratory rate with an on-screen 60-second counter, chest indrawing, stool frequency, dehydration signs; severity and Urdu action from IMCI rules config	Mobile	FE-3	Zain Ali
Test the IMCI rules against the worked cases in the WHO IMCI chart booklet	Mobile	FE-3	Zain Ali

Module 10 final

Task	Layer	Scope link	Owner
Polio campaign progress, refusals and zero-dose counts per area	Web + API	FE-1	Zain Abbas
Map layers for malnutrition hotspots and immunisation coverage gaps, beside the high-risk pregnancy layer	Web	FE-1	Zain Abbas
Full report set: adds polio progress, immunisation coverage and nutrition screening to the weekly and monthly PDF and Excel reports	API + Web	FE-2	Zain Abbas
Admin: campaign rounds and Nutrition Rehabilitation Centre records	Web + API	FE-3	Zain Abbas
Performance: database indexes, paginated tables, clustered map markers for large areas	Web + API + DB	FE-1	Zain Abbas

Exit gate for Phase 3

- ☐ A polio round runs end to end offline, including a refusal, its revisit and a zero-dose child
- ☐ A child's EPI timeline produces a defaulter alert that reaches the supervisor
- ☐ MUAC, Z-score and IMCI results match WHO reference cases
- ☐ Dashboard, map layers and reports cover all ten modules

Phase 4: Hardening and final delivery (weeks 25–28)

Phase 4 adds no new features; it proves the system works under field conditions and packages it for evaluation. Freeze features at the start of week 25 and only fix bugs after that.

Week	Task	Owner
25	End-to-end scenarios across all modules: registration → visit → Emergency → alert → referral outcome → report; polio round; EPI defaulter; SAM referral	Shared
25	Field simulation on 3 or more phones in airplane mode for several days of synthetic work, then a mass sync with deliberate conflicts	Shared
26	Low-spec device test (a 2 GB RAM Android phone): app start time, form speed, inference time, sync of 1,000 records, battery use	Zain Abbas
26	Security review: expired and tampered tokens, SQL injection attempts, HTTPS enforcement, lost-device test (local database unreadable without the password)	Zain Ali
26	Usability check of the Urdu interface and voice guidance; with LHWs only if IEC approval has arrived, otherwise with peers on dummy data	Shared
27	Bug fixing and regression run of the full test suite	Shared
27	Documentation: model report (metrics, confusion matrix, SHAP), API reference, test report, deployment guide, user guides for LHW (Urdu), supervisor and admin	Shared
28	Final mockups for Appendix A, signed release APK, staging deployment, demo script with synthetic data, presentation	Shared

Exit gate for Phase 4

- ☐ All Phase 1–3 exit gates still pass on the release build
- ☐ No open critical or high-severity bugs
- ☐ Documentation and Appendix A mockups complete

- ☐ Demo rehearsed twice on the release APK and staging server

Cross-cutting tracks

Four concerns run through every phase; each new module must follow these rules rather than invent its own.

Offline sync

Every write on the device goes to its local table and to an outbox in the same transaction. The sync service pushes outbox rows in batches (about 100 per request) when online, and the server replies with the sequence number it assigned to each UUID. Pull uses the last sequence number the device has seen, so nothing depends on device clocks.

Emergency alert records always go first in the push order (M5 FE-4). Report images upload separately after the data rows, so a slow photo never delays clinical data. Resending the same UUID is harmless; a different record for the same woman on the same day goes to the supervisor conflict queue (M3 FE-2).

Security and access

Rule	Scope link
Passwords hashed on the server with bcrypt; short-lived access token, longer refresh token, both revocable on deactivation	M1 FE-2, FE-3
Role check middleware on every route: LHW, supervisor, admin	M1 FE-2
Area scoping in every query: an LHW sees only her area, a supervisor only the areas assigned to them	M1 FE-1
Parameterised SQL only, Joi validation before any database call	Tools: Joi
Local database encrypted with AES-256; report images stored encrypted; nothing patient-related in plain shared storage	M3 FE-2, LI-8
Audit row for every create, edit, delete, referral, alert and login	M10 FE-3

Urdu and voice

Write no visible string in Dart code; every label goes into the ARB files from the start, with an Urdu and an English entry. Test every screen on a small phone, because Nastaliq text is taller than Latin text and overflows easily. For voice guidance, check whether the test phones have an Urdu text-to-speech voice in Phase 0; if not, record short audio clips for the fixed field labels (see Risks).

Testing

Level	Tool	What it covers
Unit	flutter_test, Jest, pytest	Form validation, danger-sign rules, EPI dates, MUAC and Z-score maths, IMCI rules, regex extraction, model metrics
Widget	flutter_test	Urdu screens render without overflow, RTL layout, emergency screen states
API	Jest + Supertest	Auth, role checks, area scoping, sync push and pull, conflict detection
Integration	Manual scripts on real phones	Offline-to-online flows, three alert options, multi-device sync
Model	pytest + saved metrics	Hold-out recall threshold, ONNX parity

Each FE is done only when its tests pass in CI and it works on a real phone in airplane mode.

Data model overview

The same tables exist in PostgreSQL and, for field data, in the device's Drift database. Every synced table carries the same base columns: id (UUID), server_seq, area_id, created_by, created_on_device, synced_at and deleted_at.

Table group	Key tables	Built in	Used by
Geography	districts, tehsils, union_councils, areas	Phase 0	M1, M10
Users and access	users (with role), lhw_profiles, supervisor_areas, devices, refresh_tokens, otp_codes	Phase 0-1	M1

Table group	Key tables	Built in	Used by
System	audit_log, sync_conflicts, report_jobs	Phase 0–1	M3, M10
Households and women	households (with GPS), women, pregnancies, obstetric_history	Phase 1	M2, M7
Visits	visits (all vitals and symptoms)	Phase 1	M3, M4, M6
Facilities	hospitals, referral_centres, escalation_contacts	Phase 1–2 (Nutrition Rehabilitation Centres in Phase 3)	M5, M9, M10
Risk	risk_assessments (model result, rule result, final level, explanation key, model version)	Phase 2	M4, M10
Emergency	referrals, emergency_alerts, alert_attempts (channel, status, time), alert_acknowledgements	Phase 2	M5, M10
ANC and records	anc_schedule, tt_doses, supplement_logs, health_documents (image, extracted values, confidence), trend_results	Phase 2	M6
Children	children	Phase 3	M7, M8, M9
Polio	campaigns, campaign_household_records, refusals, revisits	Phase 3	M7
Immunisation	epi_schedule (config), immunisations	Phase 3	M8
Nutrition	nutrition_screenings (MUAC, Z-scores), sam_followups, imci_assessments	Phase 3	M9

Store the model version on every risk assessment, so results stay traceable when the model is retrained on field data later (LI-2).

Risks and decisions to settle early

The first row is the most urgent: the model planned for Module 4 needs two inputs that the Module 3 form does not collect. Settle all of these in Phase 0 (task P0-11) so later phases do not stall.

Risk or decision	Why it matters	Recommended action	Settle by
Model inputs missing from the visit form	The UCI dataset uses blood sugar and heart rate; Module 3 FE-1 collects neither	Add pulse (any LHW can count it) and optional blood sugar to the form; train one model with all six features and one without blood sugar, and use the second when no reading exists	Phase 0
OTP channel	SMS OTP needs a paid gateway, which LI-4 says the prototype avoids	Use email OTP or an admin-issued one-time code for the prototype; keep the OTP service swappable	Phase 0
Urdu text in PDF reports	pdfkit may not join Nastaliq letters or order right-to-left text correctly	Test a one-page Urdu PDF in Phase 0; if it breaks, render the report as HTML with the Urdu font and print it to PDF with headless Chrome (Puppeteer)	Phase 0
Urdu voice on the phone	Many Android phones have no Urdu text-to-speech voice installed	Check the test phones; if missing, record short Urdu clips for the fixed field labels and play those instead	Phase 0
Where Layer 1 push lands	Firebase Cloud Messaging needs the supervisor's device	Add a small supervisor role in the Flutter app (alerts inbox and acknowledge) plus browser alerts on the dashboard	Phase 1

Risk or decision	Why it matters	Recommended action	Settle by
Urdu SMS length	Urdu SMS fits about 70 characters per part, so alerts split into several parts	Keep the alert template short, with patient ID and key facts first	Phase 2
SMS and call permissions	Google Play restricts apps that send SMS or place calls	Distribute the FYP build as a signed APK, not through Play Store	Phase 2
OCR language	ML Kit reads Latin script, not Urdu	Fine for English lab reports; Urdu or handwritten reports use manual entry (LI-9)	Phase 2
Phase 2 workload	Zain Ali carries both Module 5 and Module 6	Zain Abbas takes the Module 6 analytics worker (FE-3), since it is Python and he runs the ML track	Phase 1
Small training dataset	About 1,000 rows with many duplicates; scores can look better than they are	Remove duplicates, report hold-out results honestly, and present field-data retraining as future work (LI-2)	Phase 2
IEC approval delay	No real patient data before approval (LI-10)	All testing and demos on the synthetic data generator from P0-8	Ongoing

Definition of done

A module counts as finished only when every item below is true. Copy this list into each module's tracking issue on GitHub.

- ☐ Every FE in the scope document for that module works on a real Android phone
- ☐ Works fully offline where the scope says so, and syncs correctly afterwards
- ☐ All screens in Urdu with no text overflow; numeric values display left to right
- ☐ API routes validated with Joi, role-checked and area-scoped
- ☐ Every create, edit and delete writes an audit row
- ☐ Unit and API tests written and passing in CI
- ☐ Module 10 shows the module's data (dashboard, map or report, as the scope says)

- ☐ Synthetic data for the module added to the seed script
- ☐ Reviewed and merged by the other team member
- ☐ Short note added to the user guide and test report